1.1.2. Merge Sort - Divi... 28:19

You are given an array of n integers, and your task is to sort this array in non-decreasing order using the Merge Sort algorithm, which is based on the Divide and Conquer strategy.

Implement Merge Sort by defining the following user-defined functions:

- **mergeSort()** - recursively divides the array into subarrays until each subarray has a single element.
- **merge()** - merges two sorted subarrays into a single sorted array.

Input Format:

- The first line contains a single integer n representing the number of elements in the array.
- The second line contains n space-separated integers representing the elements of the array.

Output Format:

- Print the sorted array as a single line of n space-separated integers.

Constraints:

- $1 \leq n \leq 10^6$
- $-10^9 \leq Array\ Elements \leq 10^9$

Sample Test Cases +

Explorer

C merge.c

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Debugger

```c
#include <stdio.h>
#include <stdlib.h>

void merge( int arr[], int left, int mid, int right) {
      int n1 = mid - left + 1;
    int n2 = right - mid;


    //SHOURYAM GARG 25070521094
    int *L = (int *)malloc(n1 * sizeof(int));
    int *R = (int *)malloc(n2 * sizeof(int));


    for (int i = 0; i < n1; i++)
        L[i] = arr[left + i];
    for (int j = 0; j < n2; j++)
        R[j] = arr[mid + 1 + j];

      int i = 0, j = 0, k = left;
    while (i < n1 && j < n2) {
        if (L[i] <= R[j]) {
            arr[k] = L[i];
        i++;
        } else {
            arr[k] = R[j];
            j++;
        }  //SHOURYAM GARG 25070521094
        k++;
```

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```
29
30      while (i < n1) {
31          arr[k] = L[i];
32          i++;
33          k++;
34           //SHOURYAM GARG
    25070521094
35      }
36      while (j < n2) {
37      arr[k] = R[j];
38          j++;
39          k++;
40      }
41
42      // Free temporary arrays
43      free(L);
44      free(R);
45
46  }
47
48  void mergeSort( int arr[],
    int left, int right ) {
49      //write your code here..
50        if (left < right) {
51          int mid = left +
    (right - left) / 2;
52
53          mergeSort(arr,
    left, mid);
54          mergeSort(arr, mid
    + 1, right);
55   //SHOURYAM GARG 25070521094
56          merge(arr, left,
    mid, right);
57        }
58
59  }
60
61  int main() {
62      int n;
63      scanf("%d", &n);
64
```

```
    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    mergeSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }

    free(arr);
    return 0;
}   //SHOURYAM GARG 25070521094
```